Assignment 1: Profiling Codes

Group 15

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Imports

```
import time
from timeit import default_timer as timer
from functools import wraps
import numpy as np
import psutil
```

Exercise I: Profiling the Julia Set Code

Task 1.1 Calculate the Clock Granularity of different Python Timers

1. **With** `time.time()`

```
def checktick():
    M = 200
    timesfound = np.empty((M,))
    for i in range(M):
        t1 = time.time() # get timestamp from timer
        t2 = time.time() # get timestamp from timer
        while (t2 - t1) < 1e-16: # if zero then we are below clock granularity, retake timing
            t2 = time.time() # get timestamp from timer
        t1 = t2 # this is outside the loop
        timesfound[i] = t1 # record the time stamp
    minDelta = 1000000
    Delta = np.diff(timesfound) # it should be cast to int only when needed
    minDelta = Delta.min()
    return minDelta
print("time.time() clock granularity: %e s" % checktick())
```

```
time.time() clock granularity: 7.152557e-07 s
```

2. **With** `timeit`

```
def checktick():
    M = 200
    timesfound = np.empty((M,))
    for i in range(M):
        t1 = timer() # get timestamp from timer
        t2 = timer() # get timestamp from timer
        while (t2 - t1) < 1e-16: # if zero then we are below clock granularity, retake timing
            t2 = timer() # get timestamp from timer
        t1 = t2 # this is outside the loop
        timesfound[i] = t1 # record the time stamp
    minDelta = 1000000
    Delta = np.diff(timesfound) # it should be cast to int only when needed
    minDelta = Delta.min()
    return minDelta
print("timeit clock granularity: %e s" % checktick())
```

```
timeit clock granularity: 1.249718e-07 s
```

3. **With** `time.time_ns()`

```
def checktick():
    M = 200
    timesfound = np.empty((M,))
    for i in range(M):
        t1 = time.time_ns() # get timestamp from timer
        t2 = time.time_ns() # get timestamp from timer
        while (t2 - t1) < 1e-16: # if zero then we are below clock granularity, retake timing
            t2 = time.time_ns() # get timestamp from timer
        t1 = t2 # this is outside the loop
```

```
        timesfound[i] = t1 # record the time stamp
    minDelta = 1000000
    Delta = np.diff(timesfound) # it should be cast to int only when needed
    minDelta = Delta.min()
    return minDelta
print("time.time() clock granularity: %e ns" % checktick())
```

```
time.time() clock granularity: 7.680000e+02 ns
```

Task 1.2 Timing the Julia set code functions

`Timeit` has the best clock granularity among the three timers tested above. Therefore, I use `timeit` to time the Julia set code functions.

```
"""Julia set generator without optional PIL-based image drawing"""
# area of complex space to investigate
x1, x2, y1, y2 = -1.8, 1.8, -1.8, 1.8
c_real, c_imag = -0.62772, -.42193

# decorator to time
def timefn(fn):
    @wraps(fn)
    def measure_time(*args, **kwargs):
        t1 = timer()
        result = fn(*args, **kwargs)
        t2 = timer()
        print(f"@timefn: {fn.__name__} took {t2 - t1} seconds")
        return result
    return measure_time

@timefn
def calc_pure_python(desired_width, max_iterations):
    """Create a list of complex coordinates (zs) and complex parameters (cs),
    build Julia set"""
    x_step = (x2 - x1) / desired_width
    y_step = (y1 - y2) / desired_width
    x = []
    y = []
    ycoord = y2
    while ycoord > y1:
        y.append(ycoord)
        ycoord += y_step
    xcoord = x1
    while xcoord < x2:
        x.append(xcoord)
        xcoord += x_step
    # build a list of coordinates and the initial condition for each cell.
    # Note that our initial condition is a constant and could easily be removed,
    # we use it to simulate a real-world scenario with several inputs to our
    # function
    zs = []
    cs = []
    for ycoord in y:
        for xcoord in x:
            zs.append(complex(xcoord, ycoord))
            cs.append(complex(c_real, c_imag))

    print("Length of x:", len(x))
    print("Total elements:", len(zs))
    start_time = time.time()
    output = calculate_z_serial_purepython(max_iterations, zs, cs)
    end_time = time.time()
    secs = end_time - start_time
    # print(calculate_z_serial_purepython.__name__ + " took", secs, "seconds")

    # This sum is expected for a 1000^2 grid with 300 iterations
    # It ensures that our code evolves exactly as we'd intended
    assert sum(output) == 33219980

@timefn
def calculate_z_serial_purepython(maxiter, zs, cs):
    """Calculate output list using Julia update rule"""
    output = [0] * len(zs)
    for i in range(len(zs)):
        n = 0
        z = zs[i]
```

```
        c = cs[i]
        while abs(z) < 2 and n < maxiter:
            z = z * z + c
            n += 1
        output[i] = n
    return output

if __name__ == "__main__":
    # Calculate the Julia set using a pure Python solution with
    # reasonable defaults for a laptop
    calc_pure_python(desired_width=1000, max_iterations=300)
```

```
Length of x: 1000
Total elements: 1000000
@timefn: calculate_z_serial_purepython took 1.8548854169785045 seconds
@timefn: calc_pure_python took 1.9846684589865617 seconds
```

The above code was run 10 times and the average time and their corresponding standard deviations are as follows:

Run	calc_pure_python (s)	calculate_z_serial_purepython (s)
1	1.6460210420191288	1.5471606670180336
2	1.829247374960687	1.7324306250084192
3	1.6537152499658987	1.5582562089548446
4	1.9431911670253612	1.846423249982763
5	1.8339538750005886	1.5811095830285922
6	1.644327334011905	1.5485797080327757
7	1.664894582994748	1.5704846250009723
8	1.64977145800367	1.5531713340315036
9	1.6521273750113323	1.5556108329910785
10	1.6534901250270195	1.5559448749991134
Average Time (s)	1.717073958402	1.6049171709048
Standard Deviation (s)	0.10355721746071	0.096193175360614

When comparing the standard deviations of the two functions with the clock granularity of `timeit`, it is obvious that they are significantly larger than the clock granularity. This indicates that the timing measurements are not significantly affected by the limitations of the timer's resolution.

Task 1.3 Profile the Julia set code with cProfile and line_profiler the computation

With cProfile

```
python -m cProfile -s cumulative JuliaSet.py
```

ncalls	tottime	percall	cumtime	percall	filename:lineno(function)
1	0.219	0.219	4.187	4.187	JuliaSet.py:24(calc_pure_python)
1	2.645	2.645	3.892	3.892	JuliaSet.py:63(calculate_z_serial_purepython)

With line_profiler

```
python -m kernprof -l JuliaSet.py
```

`Timer unit: 1e-06 s`

Function: calc_pure_python at line 23

```
Total time: 26.904 s
Line #      Hits         Time  Per Hit   % Time  Line Contents
==============================================================
    23                                           @profile
    24                                           def calc_pure_python(desired_width, max_iterations):
    25                                               """Create a list of complex coordinates (zs) and complex parameters (cs),
    26                                               build Julia set"""
    27         1          1.0      1.0      0.0      x_step = (x2 - x1) / desired_width
    28         1          1.0      1.0      0.0      y_step = (y1 - y2) / desired_width
    29         1          0.0      0.0      0.0      x = []
    30         1          1.0      1.0      0.0      y = []
    31         1          0.0      0.0      0.0      ycoord = y2
```

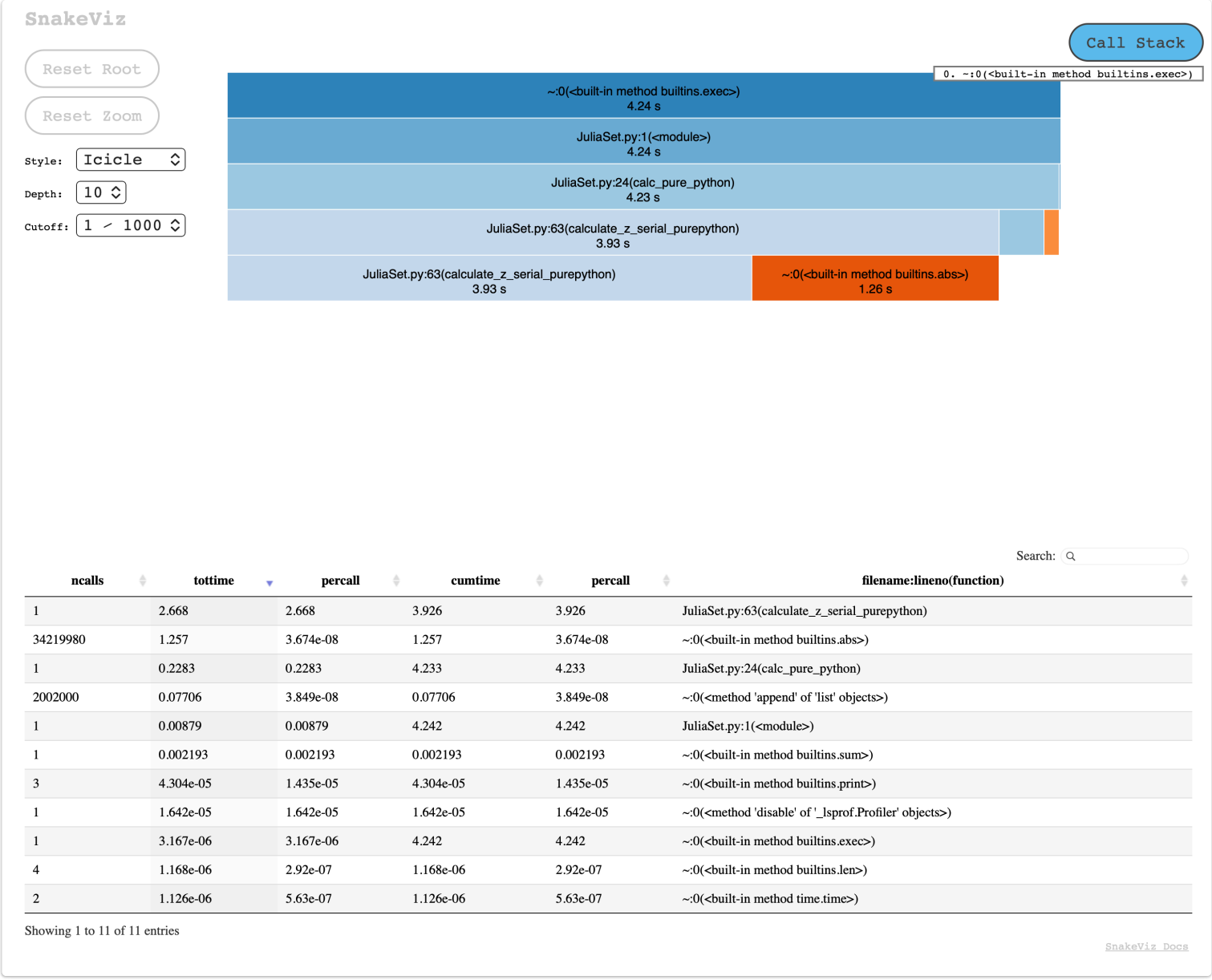
```
    32      1001        148.0      0.1      0.0      while ycoord > y1:
    33      1000        154.0      0.2      0.0          y.append(ycoord)
    34      1000        135.0      0.1      0.0          ycoord += y_step
    35         1          0.0      0.0      0.0      xcoord = x1
    36      1001        166.0      0.2      0.0      while xcoord < x2:
    37      1000        136.0      0.1      0.0          x.append(xcoord)
    38      1000        132.0      0.1      0.0          xcoord += x_step
    39                                               # build a list of coordinates and the initial condition for each cell.
    40                                               # Note that our initial condition is a constant and could easily be removed,
    41                                               # we use it to simulate a real-world scenario with several inputs to our
    42                                               # function
    43         1          0.0      0.0      0.0      zs = []
    44         1          0.0      0.0      0.0      cs = []
    45      1001        175.0      0.2      0.0      for ycoord in y:
    46   1001000     151641.0      0.2      0.6          for xcoord in x:
    47   1000000     178805.0      0.2      0.7              zs.append(complex(xcoord, ycoord))
    48   1000000     187474.0      0.2      0.7              cs.append(complex(c_real, c_imag))
    49
    50         1         21.0     21.0      0.0      print("Length of x:", len(x))
    51         1          3.0      3.0      0.0      print("Total elements:", len(zs))
    52         1          1.0      1.0      0.0      start_time = time.time()
    53         1   26382746.0 2.64e+07     98.1      output = calculate_z_serial_purepython(max_iterations, zs, cs)
    54         1          4.0      4.0      0.0      end_time = time.time()
    55         1          3.0      3.0      0.0      secs = end_time - start_time
    56         1         31.0     31.0      0.0      print(calculate_z_serial_purepython.__name__ + " took", secs, "seconds")
    57
    58                                               # This sum is expected for a 1000^2 grid with 300 iterations
    59                                               # It ensures that our code evolves exactly as we'd intended
    60         1       2212.0   2212.0      0.0      assert sum(output) == 33219980
```

Function: calculate_z_serial_purepython at line 62

```
Total time: 18.855 s

Line #      Hits         Time  Per Hit   % Time  Line Contents
==============================================================
    62                                           @profile
    63                                           def calculate_z_serial_purepython(maxiter, zs, cs):
    64                                               """Calculate output list using Julia update rule"""
    65         1        472.0    472.0      0.0      output = [0] * len(zs)
    66   1000001     176330.0      0.2      0.9      for i in range(len(zs)):
    67   1000000     163153.0      0.2      0.9          n = 0
    68   1000000     175652.0      0.2      0.9          z = zs[i]
    69   1000000     165613.0      0.2      0.9          c = cs[i]
    70  34219980    6494024.0      0.2     34.4          while abs(z) < 2 and n < maxiter:
    71  33219980    6118674.0      0.2     32.5              z = z * z + c
    72  33219980    5394531.0      0.2     28.6              n += 1
    73   1000000     166519.0      0.2      0.9          output[i] = n
    74         1          2.0      2.0      0.0      return output
```

Snakeviz Visualization

```
python -m snakeviz profile.stats --server
```



Overhead Analysis

To measure the overhead introduced by cProfile and line_profiler, the code is timed with the unix `time` command in three different scenarios: without any profiling (no @profile), with cProfile (no @profile), and with line_profiler (with decorators). Given the significant performance penalty associated with profiling, a coarse-grained timer serves as an appropriate tool for basic timing.

Scenario	Real Time (s)
Without Profiling `time python JuliaSet.py`	1.692
With cProfile `time python -m cProfile -s cumulative JuliaSet.py`	4.272
With line_profiler `time python -m kernprof -l JuliaSet.py`	29.361

Task 1.4 Memory-profile the JuliaSet code

```
python -m memory_profiler JuliaSet.py
```

> *To reduce running time, the above memory profiling was done with 10 iterations instead of 300.*

Function: calc_pure_python at line 23

```
Line #    Mem usage    Increment  Occurrences   Line Contents
=============================================================
    23   82.859 MiB   82.859 MiB           1   @profile
    24                                         def calc_pure_python(desired_width, max_iterations):
    25                                             """Create a list of complex coordinates (zs) and complex parameters (cs),
    26                                             build Julia set"""
    27   82.859 MiB    0.000 MiB           1       x_step = (x2 - x1) / desired_width
    28   82.859 MiB    0.000 MiB           1       y_step = (y1 - y2) / desired_width
    29   82.859 MiB    0.000 MiB           1       x = []
    30   82.859 MiB    0.000 MiB           1       y = []
    31   82.859 MiB    0.000 MiB           1       ycoord = y2
    32   82.906 MiB    0.031 MiB        1001       while ycoord > y1:
```

```
    33   82.906 MiB    0.016 MiB        1000           y.append(ycoord)
    34   82.906 MiB    0.000 MiB        1000           ycoord += y_step
    35   82.906 MiB    0.000 MiB           1       xcoord = x1
    36   82.938 MiB    0.031 MiB        1001       while xcoord < x2:
    37   82.938 MiB    0.000 MiB        1000           x.append(xcoord)
    38   82.938 MiB    0.000 MiB        1000           xcoord += x_step
    39                                             # build a list of coordinates and the initial condition for each cell.
    40                                             # Note that our initial condition is a constant and could easily be removed,
    41                                             # we use it to simulate a real-world scenario with several inputs to our
    42                                             # function
    43   82.938 MiB    0.000 MiB           1       zs = []
    44   82.938 MiB    0.000 MiB           1       cs = []
    45  166.688 MiB    0.000 MiB        1001       for ycoord in y:
    46  166.688 MiB    0.000 MiB     1001000           for xcoord in x:
    47  166.688 MiB   72.625 MiB     1000000               zs.append(complex(xcoord, ycoord))
    48  166.688 MiB   11.125 MiB     1000000               cs.append(complex(c_real, c_imag))
    49
    50  166.688 MiB    0.000 MiB           1       print("Length of x:", len(x))
    51  166.688 MiB    0.000 MiB           1       print("Total elements:", len(zs))
    52  166.688 MiB    0.000 MiB           1       start_time = time.time()
    53  188.656 MiB  188.656 MiB           1       output = calculate_z_serial_purepython(max_iterations, zs, cs)
    54  188.656 MiB    0.000 MiB           1       end_time = time.time()
    55  188.656 MiB    0.000 MiB           1       secs = end_time - start_time
    56  188.656 MiB    0.000 MiB           1       print(calculate_z_serial_purepython.__name__ + " took", secs, "seconds")
    57
    58                                             # This sum is expected for a 1000^2 grid with 300 iterations
    59                                             # It ensures that our code evolves exactly as we'd intended
    60                                             # assert sum(output) == 33219980
```

Function: calculate_z_serial_purepython at line 62

```
Line #    Mem usage    Increment  Occurrences   Line Contents
=============================================================
    62  166.688 MiB  166.688 MiB           1   @profile
    63                                         def calculate_z_serial_purepython(maxiter, zs, cs):
    64                                             """Calculate output list using Julia update rule"""
    65  174.344 MiB    7.656 MiB           1       output = [0] * len(zs)
    66  188.656 MiB    0.000 MiB     1000001       for i in range(len(zs)):
    67  188.656 MiB    0.000 MiB     1000000           n = 0
    68  188.656 MiB    7.156 MiB     1000000           z = zs[i]
    69  188.656 MiB    7.156 MiB     1000000           c = cs[i]
    70  188.656 MiB    0.000 MiB     3988736           while abs(z) < 2 and n < maxiter:
    71  188.656 MiB    0.000 MiB     2988736               z = z * z + c
    72  188.656 MiB    0.000 MiB     2988736               n += 1
    73  188.656 MiB    0.000 MiB     1000000           output[i] = n
    74  188.656 MiB    0.000 MiB           1       return output
```

> *The following memory statistics were collected with the original 300 iterations.*

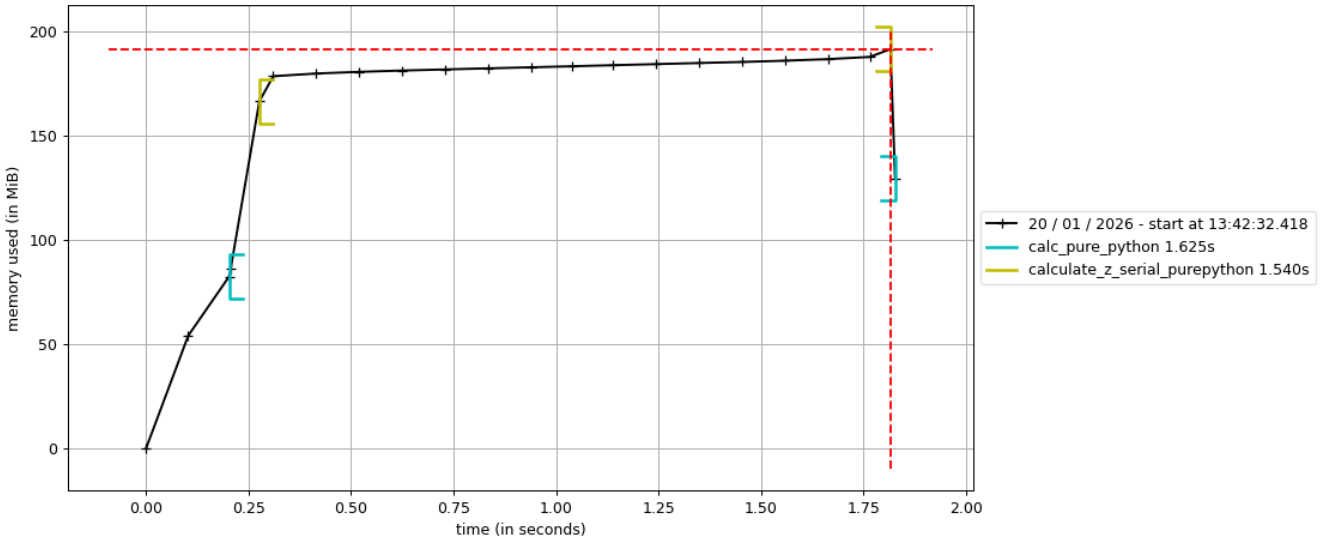
Collecting Memory Statistics

```
python -m mprof run JuliaSet.py
```

Visualising Memory Usage with mprof

```
python -m mprof plot julia_mprofile.dat
```

/Library/CloudStorage/SynologyDrive-Home/Drive/Documents/University/CUHK/Learning Materials/Sync/DD2358/asm1/.venv/bin/python' JuliaSet.py



Overhead Analysis

To measure the overhead introduced by cProfile and line_profiler, the code is timed with the unix `time` command in three different scenarios: without any profiling (no @profile), with memory_profiler (with @profile), and with mprof (with @profile). Given the significant performance penalty associated with profiling, a coarse-grained timer serves as an appropriate tool for basic timing. **All are done with 10 iterations instead of 300 to reduce running time.**

Scenario	Real Time (s)
Without Profiling `time python JuliaSet.py`	0.308
With memory_profiler `time python -m memory_profiler JuliaSet.py`	3:32.69
With mprof `time python -m mprof run JuliaSet.py`	0.715

Exercise II: Profiling Diffusion Process Code

The file `Diffusion.py` contains the following code for profiling with the experiment being run for 300 iterations.

```
grid_shape = (640, 640)

def evolve(grid, dt, D=1.0):
    xmax, ymax = grid_shape
    new_grid = [[0.0] * ymax for x in range(xmax)]
    for i in range(xmax):
        for j in range(ymax):
            grid_xx = (
                    grid[(i + 1) % xmax][j] + grid[(i - 1) % xmax][j] - 2.0 * grid[i][j]
            )
            grid_yy = (
                    grid[i][(j + 1) % ymax] + grid[i][(j - 1) % ymax] - 2.0 * grid[i][j]
            )
            new_grid[i][j] = grid[i][j] + D * (grid_xx + grid_yy) * dt
    return new_grid

def run_experiment(num_iterations):
    # Setting up initial conditions
    xmax, ymax = grid_shape
    grid = [[0.0] * ymax for x in range(xmax)]

    # These initial conditions are simulating a drop of dye in the middle of our
    # simulated region
    block_low = int(grid_shape[0] * 0.4)
    block_high = int(grid_shape[0] * 0.5)
    for i in range(block_low, block_high):
        for j in range(block_low, block_high):
            grid[i][j] = 0.005

    # Evolve the initial conditions
    for i in range(num_iterations):
        grid = evolve(grid, 0.1)
```

```
if __name__ == "__main__":
    run_experiment(300)
```

Task 2.1 Profile the diffusion code with cProfile and line_profiler the computation

With cProfile

```
python -m cProfile -s cumulative Diffusion.py
```

```
304 function calls in 18.929 seconds

ncalls  tottime  percall  cumtime  percall filename:lineno(function)
     1    0.000    0.000   18.929   18.929 {built-in method builtins.exec}
     1    0.002    0.002   18.929   18.929 Diffusion.py:1(<module>)
     1    0.467    0.467   18.927   18.927 Diffusion.py:17(run_experiment)
   300   18.459    0.062   18.459    0.062 Diffusion.py:3(evolve)
     1    0.000    0.000    0.000    0.000 {method 'disable' of '_lsprof.Profiler' objects}
```

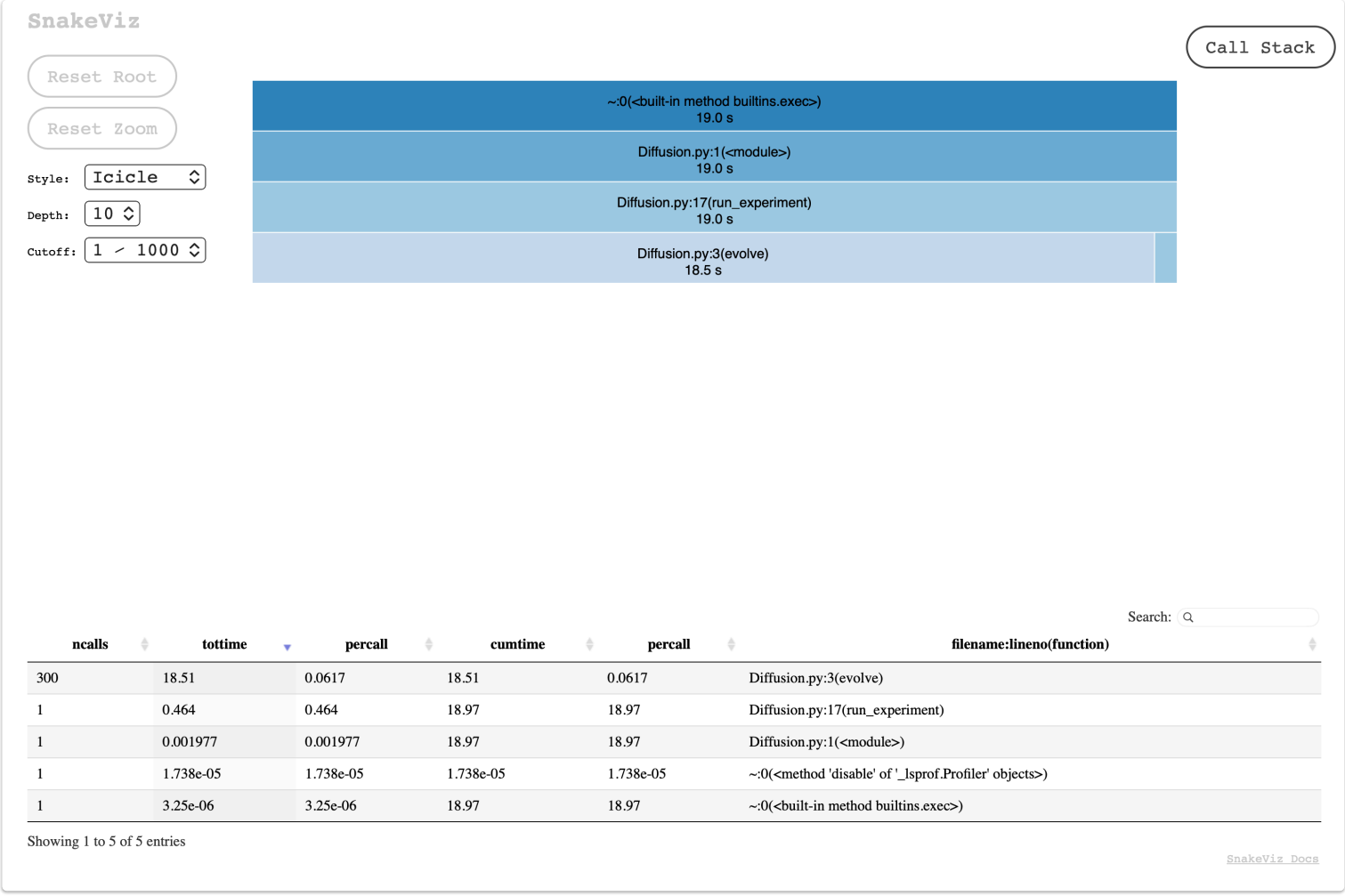
With line_profiler

```
python -m kernprof -l Diffusion.py
```

```
Timer unit: 1e-06 s

Total time: 140.727 s
File: Diffusion.py
Function: evolve at line 3

Line #      Hits         Time  Per Hit   % Time  Line Contents
==============================================================
     3                                           @profile
     4                                           def evolve(grid, dt, D=1.0):
     5       300        294.0      1.0      0.0      xmax, ymax = grid_shape
     6       300      37253.0    124.2      0.0      new_grid = [[0.0] * ymax for x in range(xmax)]
     7    192300      33973.0      0.2      0.0      for i in range(xmax):
     8 123072000   21260026.0      0.2     15.1          for j in range(ymax):
     9 122880000   20142275.0      0.2     14.3              grid_xx = (
    10 122880000   27531128.0      0.2     19.6                      grid[(i + 1) % xmax][j] + grid[(i - 1) % xmax][j] - 2.0 * grid[i][j]
    11                                                       )
    12 122880000   20100690.0      0.2     14.3              grid_yy = (
    13 122880000   27626248.0      0.2     19.6                      grid[i][(j + 1) % ymax] + grid[i][(j - 1) % ymax] - 2.0 * grid[i][j]
    14                                                       )
    15 122880000   23994958.0      0.2     17.1              new_grid[i][j] = grid[i][j] + D * (grid_xx + grid_yy) * dt
    16       300        521.0      1.7      0.0      return new_grid
```

Snakeviz Visualization

```
python -m snakeviz diffusion_profile.stats --server
```



Task 2.2 Memory-profile the diffusion code

```
python -m memory_profiler Diffusion.py
```

> *To reduce running time, the above memory profiling was done with 5 iterations instead of 300 for the `evolve` function.*

```
Line #    Mem usage    Increment  Occurrences   Line Contents
=============================================================
     3   92.359 MiB   42.969 MiB           5   @profile
     4                                         def evolve(grid, dt, D=1.0):
     5   92.359 MiB    0.000 MiB           5       xmax, ymax = grid_shape
     6   92.359 MiB    3.297 MiB        3205       new_grid = [[0.0] * ymax for x in range(xmax)]
     7  102.359 MiB -11521.172 MiB        3205       for i in range(xmax):
     8  102.359 MiB -7394948.047 MiB     2051200           for j in range(ymax):
     9  102.359 MiB -7383370.781 MiB     2048000               grid_xx = (
    10  102.359 MiB -7383426.875 MiB     2048000                       grid[(i + 1) % xmax][j] + grid[(i - 1) % xmax][j] - 2.0 * grid[i]
[j]
    11                                                           )
    12  102.359 MiB -7383395.875 MiB     2048000               grid_yy = (
    13  102.359 MiB -7383395.875 MiB     2048000                       grid[i][(j + 1) % ymax] + grid[i][(j - 1) % ymax] - 2.0 * grid[i]
[j]
    14                                                           )
    15  102.359 MiB -7383395.875 MiB     2048000               new_grid[i][j] = grid[i][j] + D * (grid_xx + grid_yy) * dt
    16  102.359 MiB    0.000 MiB           5       return new_grid
```

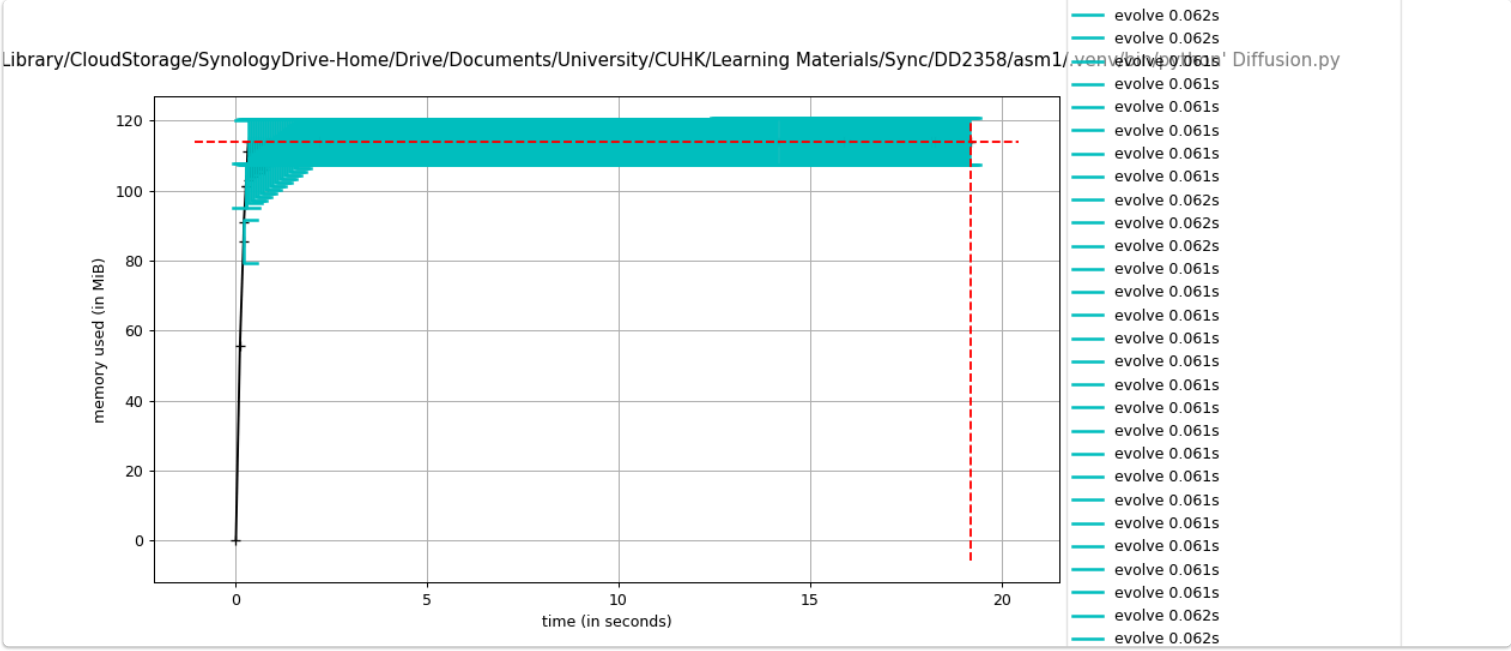
> *The following memory statistics were collected with the original 300 iterations.*

Collecting Memory Statistics

```
python -m mprof run Diffusion.py
```

Visualising Memory Usage with mprof

```
python -m mprof plot diffusion_mprofile.dat
```



Bonus Exercise: Develop your profiler tool for monitoring CPU percentage use with psutil

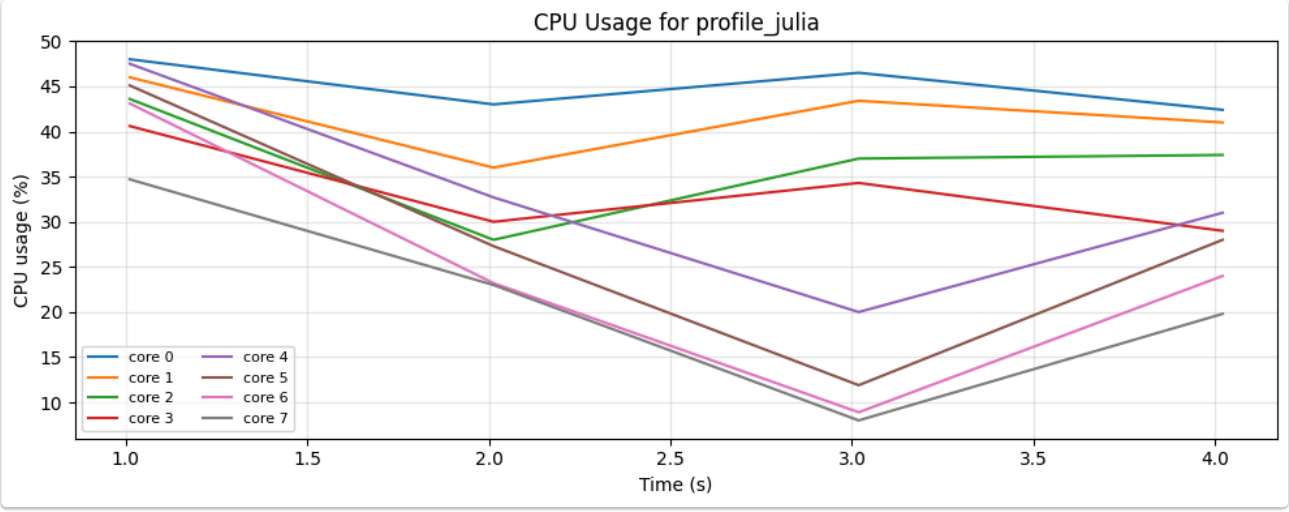
The profiler implemented below uses a `CpuProfiler` class that runs a background thread to periodically sample CPU usage statistics for each core using the `psutil` library. It records these samples along with corresponding timestamps while a target function executes. A decorator cpu_profile is provided as a trigger for profiling of the target function. Once profiling is complete, the collected CPU usage data can be visualized over time using matplotlib, allowing users to monitor and analyze CPU utilization during code execution.

```
import time
import threading
import matplotlib.pyplot as plt

# Sample CPU usage while the function runs, then show ONE final plot + table
class CpuProfiler:
    def __init__(self, interval=1.0):
        self.interval = interval
        self.timestamps = []
        self.samples = []
        self._stop = threading.Event()
        self._thread = None

# Set up a function that runs in a background thread and collects CPU usage data and timestamps
    def _collect(self):
        start = time.time()
        while not self._stop.is_set():
            values = psutil.cpu_percent(interval=self.interval, percpu=True)
            self.timestamps.append(time.time() - start)
            self.samples.append(values)

# Initialisation
    def start(self):
        self.timestamps = []
        self.samples = []
        self._stop.clear()
        psutil.cpu_percent(interval=1, percpu=True)  # prime psutil
        self._thread = threading.Thread(target=self._collect, daemon=True)
        self._thread.start()

# Stop the background thread
    def stop(self):
        self._stop.set()
        if self._thread is not None:
            self._thread.join()

# Plot the final CPU usage using matplotlib
    def plot_final(self, title="CPU Usage per Core"):
        if not self.samples:
            print("No samples collected.")
            return
        plt.figure(figsize=(10, 4))
        num_cores = len(self.samples[0])
        for core_idx in range(num_cores):
            y = [s[core_idx] for s in self.samples]
            plt.plot(self.timestamps, y, label=f"core {core_idx}")
```

```python
        plt.xlabel("Time (s)")
        plt.ylabel("CPU usage (%)")
        plt.title(title)
        plt.legend(ncol=2, fontsize=8)
        plt.grid(True, alpha=0.3)
        plt.tight_layout()
        plt.show()

# Print the CPU usage in a table format
    def table(self):
        if not self.samples:
            print("No samples collected.")
            return
        headers = ["time(s)"] + [f"Core {i}" for i in range(len(self.samples[0]))]
        print("| " + " | ".join(headers) + " |")
        print("| " + " | ".join(["---"] * len(headers)) + " |")
        for t, row in zip(self.timestamps, self.samples):
            values = [f"{t:.1f}"] + [f"{v:.1f}" for v in row]
            print("| " + " | ".join(values) + " |")

# A decorator to profile the CPU usage of a function
def cpu_profile(interval=1.0, post_seconds=0.0):
    def decorator(func):
        def wrapper(*args, **kwargs):
            profiler = CpuProfiler(interval=interval)
            profiler.start()
            try:
                result = func(*args, **kwargs)
            finally:
                if post_seconds > 0:
                    time.sleep(post_seconds)
                profiler.stop()
            profiler.plot_final(title=f"CPU Usage for {func.__name__}")
            profiler.table()
            return result
        return wrapper
    return decorator
```

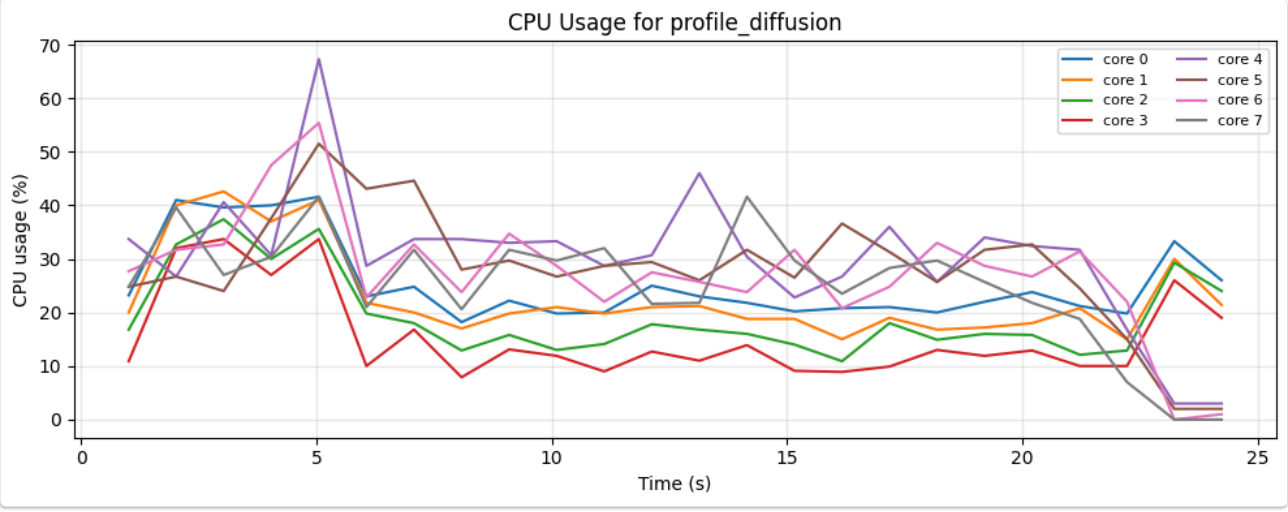
```python
julia_width = 1000
julia_iters = 300
diffusion_iters = 300

@cpu_profile(interval=1.0, post_seconds=2)
def profile_julia():
    return calc_pure_python(desired_width=julia_width, max_iterations=julia_iters)

@cpu_profile(interval=1.0, post_seconds=2)
def profile_diffusion():
    return run_experiment(diffusion_iters)

print("Profiling Julia...")
profile_julia()

print("Profiling diffusion...")
profile_diffusion()
```

```
Profiling Julia... (this may take a while)
Length of x: 1000
Total elements: 1000000
@timefn: calculate_z_serial_purepython took 1.8221291250083596 seconds
@timefn: calc_pure_python took 1.9617605420062318 seconds
```



time(s)	Core 0	Core 1	Core 2	Core 3	Core 4	Core 5	Core 6	Core 7
1.0	48.0	46.0	43.6	40.6	47.5	45.1	43.1	34.7
2.0	43.0	36.0	28.0	30.0	32.7	27.3	23.2	23.0
3.0	46.5	43.4	37.0	34.3	20.0	11.9	8.9	8.0
4.0	42.4	41.0	37.4	29.0	31.0	28.0	24.0	19.8

Profiling diffusion... (this may take a while)



time(s)	Core 0	Core 1	Core 2	Core 3	Core 4	Core 5	Core 6	Core 7
1.0	23.2	20.0	16.8	10.9	33.7	24.8	27.7	24.8
2.0	41.0	40.0	32.7	32.0	26.7	26.7	31.7	39.6
3.0	39.6	42.6	37.4	33.7	40.6	24.0	32.7	27.0
4.0	40.0	37.0	30.0	27.0	30.7	37.6	47.5	30.4
5.0	41.6	41.0	35.6	33.7	67.3	51.5	55.4	41.6
6.1	23.0	21.8	19.8	10.0	28.7	43.1	22.8	21.0
7.1	24.8	20.0	18.0	16.8	33.7	44.6	32.7	31.7
8.1	18.2	17.0	12.9	7.9	33.7	28.0	23.8	20.6
9.1	22.2	19.8	15.8	13.1	33.0	29.7	34.7	31.7
10.1	19.8	21.0	13.0	11.9	33.3	26.7	28.7	29.7
11.1	20.0	19.8	14.1	9.0	28.7	28.7	22.0	32.0
12.1	25.0	21.0	17.8	12.7	30.7	29.4	27.5	21.6
13.1	23.0	21.2	16.8	11.0	46.0	26.0	25.7	21.8
14.1	21.8	18.8	16.0	13.9	30.4	31.7	23.8	41.6
15.2	20.2	18.8	14.0	9.1	22.8	26.5	31.7	29.7
16.2	20.8	15.0	10.9	8.9	26.7	36.6	20.8	23.5
17.2	21.0	19.0	18.0	9.9	36.0	31.3	24.8	28.3
18.2	20.0	16.8	14.9	13.0	25.7	25.7	33.0	29.7
19.2	22.0	17.2	16.0	11.9	34.0	31.7	28.7	25.7
20.2	23.8	18.0	15.8	12.9	32.4	32.7	26.7	21.8
21.2	21.2	20.8	12.1	10.0	31.7	24.5	31.4	18.8
22.2	19.8	15.0	12.9	10.0	16.8	15.0	22.0	7.0
23.2	33.3	30.0	29.3	26.0	3.0	2.0	0.0	0.0
24.2	26.0	21.4	24.0	19.0	3.0	2.0	1.0	0.0